M Afaq Bhatti

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SUMMARY

AI/ML Engineer and CS student with production experience in end-to-end ML pipelines, multi-agent systems, and cloud-deployed inference engines. Python, LangGraph, AWS, MLOps. Kaggle Global Top-14 Data Scientist.

EDUCATION

COMSATS University Islamabad (CUI) *Islamabad, PK*
BS Computer Science — CGPA: 3.4/4.0 | Coursework: ML, Gen AI, Agentic AI, Deep Learning, DevOps, MLOps, DSA *Feb 2023 – Present*

WORK EXPERIENCE

ITSOLERA *Islamabad, PK*
Data Science Intern *Jul 2025 – Sep 2025*

- Led a team to architect and deploy **6 production-grade ML solutions** — traffic accident forecasting and demographic prediction models.
- Built **FinSentRisk**, a financial intelligence platform using LSTM/ARIMA time-series forecasting for market trend prediction and risk assessment.
- Applied full **MLOps stack**: DVC data versioning, MLflow experiment tracking, GitHub Actions CI/CD, automated model retraining.
- Reduced model deployment time by **40%** via containerized Flask inference APIs and automated data ingestion pipelines.

PROJECTS

FinOps-Optimized GraphRAG: Local Knowledge Graph Engine *Apr 2026*

- Engineered a **GraphRAG** system using **LightRAG** + **Gemini 2.5 Flash** for entity-relationship extraction, cutting inference cost to **1/6000th** of traditional RAG.
- Multi-mode retrieval (hybrid/global/local) combining **NanoVectorDB** semantic search with graph traversal; deployed on **AWS EC2** + **S3** via **FastAPI**.

Distributed Multi-Agent Market Intelligence System *Mar 2026*

- Built a **parallel multi-agent system** (LangGraph + Groq LLaMA-3.3): 5 specialist agents analyze price action, sentiment, on-chain metrics, macro, and risk simultaneously.
- **Fan-out → fan-in architecture** synthesizes agent outputs into a unified trading verdict, cutting execution time **5x** vs. sequential; deployed on **AWS EC2** with Streamlit dashboard.

VisionRAG: Multimodal Document Intelligence *Apr 2026*

- Visual RAG pipeline using **ColPali (ColQwen2)** + **Gemini** to query complex PDFs (tables/charts) without OCR data loss; sub-second responses on multi-page corpora.

End-to-End MLOps Emotion Detection Pipeline *Jul 2025*

- CNN-based emotion detection served via Flask; full MLOps pipeline — **DVC**, **MLflow**, **GitHub Actions** CI/CD — enabling zero-touch model retraining on new data.

TECHNICAL SKILLS

Languages: Python, C++, SQL, Bash
AI / ML: TensorFlow, PyTorch, Scikit-learn, OpenCV, LangChain, LangGraph, LangSmith, LightRAG
Generative AI & LLMs: Prompt Engineering, RAG, GraphRAG, Multi-Agent Systems, LLaMA, Gemini, Groq API, Agentic AI
MLOps & DevOps: MLflow, DVC, Docker, GitHub Actions (CI/CD), Jenkins, Kubernetes, Git
Cloud & APIs: AWS EC2, S3, IAM, FastAPI, Flask, Streamlit, REST API Design, Microservices, WebSockets
Databases: NumPy, Pandas, MongoDB, MySQL, PostgreSQL, NanoVectorDB

ACHIEVEMENTS & CERTIFICATIONS

- **Kaggle Global Top 14 Data Scientist** — Top 14 worldwide out of thousands; mastery in ML, feature engineering, and predictive modeling.
- **Certifications:** Complete Data Science, ML, DL & NLP Bootcamp (Krish Naik, Udemy) | Advanced SQL (Kaggle) | Geospatial Analysis (Kaggle) | Computer Vision (Udemy) | Intro to Game AI & RL (Kaggle)